Cosine Similarity

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Frobenius norm

The Frobenius norm of a matrix A is defined as

$$||A||_F = \sqrt{\sum_{i,j} A_{ij}^2}$$

where A_{ij} are the elements of the matrix A, and the sum is taken over all rows i and columns j of the matrix.

For example, if we have a matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

The Frobenius norm of A would be calculated as:

$$||A||_F = \sqrt{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2} = \sqrt{1 + 4 + 9 + 16 + 25 + 36} = \sqrt{91}$$

The interpretation of the Frobenius norm is the Euclidean length of the matrix when it is flattened into one long vector. In the case of our matrix $A \in \mathbb{R}^{3\times 2}$, it is literally the distance from the origin to the point in 6-dimensional space $(3 \cdot 2)$ representing the magnitude of the elements of the matrix. It measures the overall magnitude of the matrix.

Frobenius inner product

The Frobenius inner product of two matrices A and B of the same dimensions is defined as:

$$\langle A, B \rangle_F = \sum_{i,j} A_{ij} B_{ij} = \text{tr}\ (A^T B)$$

This is essentially the sum of the products of corresponding elements from the two matrices. It can be thought of as a generalization of the dot product for vectors to matrices.

For example, if we have two matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = [7 \;\; 8 \;\; 9]$$

We can reshape B to match the dimensions of A (if necessary) by embedding into a 3x2 matrix, for example:

$$B' = \begin{bmatrix} 7 & 0 \\ 8 & 0 \\ 9 & 0 \end{bmatrix}$$

$\sum_{i,j} A_{ij} B'_{ij}$ can be rewritten as $\sum A \circ B'$, which is the sum of the Hadamard product of A and B'. The Hadamard product is the element-wise product of two matrices of the same dimensions. In this case, it would be calculated as:

$$A \circ B' = \begin{bmatrix} 1 \cdot 7 & 2 \cdot 0 \\ 3 \cdot 8 & 4 \cdot 0 \\ 5 \cdot 9 & 6 \cdot 0 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 24 & 0 \\ 45 & 0 \end{bmatrix}$$

Then the Frobenius inner product would be calculated as:

$$\langle A, B' \rangle_F = \sum A \circ B' = 7 + 0 + 24 + 0 + 45 + 0 = 76$$

To explore how the Frobenius inner product can be expressed as the trace of the product of the transpose of A and B, we can compute $A^T B$:

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

$$A^T B' = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 8 & 0 \\ 9 & 0 \end{bmatrix} = \begin{bmatrix} (1 \cdot 7 + 3 \cdot 8 + 5 \cdot 9) & (1 \cdot 0 + 3 \cdot 0 + 5 \cdot 0) \\ (2 \cdot 7 + 4 \cdot 8 + 6 \cdot 9) & (2 \cdot 0 + 4 \cdot 0 + 6 \cdot 0) \end{bmatrix} = \begin{bmatrix} 76 & 0 \\ (14 + 32 + 54) & 0 \end{bmatrix} = \begin{bmatrix} 76 & 0 \\ (100) & 0 \end{bmatrix}$$

The trace of $A^T B'$ is the sum of the diagonal elements:

$$\text{tr}(A^T B') = 76 + 0 = 76$$

Thus, the Frobenius inner product $\langle A, B' \rangle_F$ can indeed be expressed as the trace of the product of the transpose of A and B', confirming that:

$$\langle A, B' \rangle_F = \text{tr}(A^T B')$$

The interpretation of the Frobenius inner product is the alignment of the matrices. The more dimensions that are aligned the larger the inner product. If the matrices are orthogonal, the inner product is zero. If they are perfectly aligned, the inner product is maximized. If they are opposite, the inner product is negative.

Cosine similarity

Reviewing the law of cosines, for two vectors u and v in $\mathbb{R}^n$, the cosine of the angle θ between them is given by:

$$\cos(\theta) = \frac{u \cdot v}{||u|| \cdot ||v||}$$

The cosine similarity between two matrices A and B in $\mathbb{R}^{m \times n}$ can be defined analogously using the Frobenius inner product and Frobenius norms:

$$\cos(\theta) = \frac{\langle A, B \rangle_F}{||A||_F \cdot ||B||_F}$$

For example, if we have two matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 0 \\ 8 & 0 \\ 9 & 0 \end{bmatrix}$$

We can calculate the cosine similarity as follows:

1. Calculate the Frobenius inner product $\langle A, B \rangle_F$:

$$\langle A, B \rangle_F = 76$$

2. Calculate the Frobenius norms $||A||_F$ and $||B||_F$:

$$||A||_F = \sqrt{91}$$

$$||B||_F = \sqrt{7^2 + 0^2 + 8^2 + 0^2 + 9^2 + 0^2} = \sqrt{49 + 0 + 64 + 0 + 81 + 0} = \sqrt{194}$$

3. Substitute these values into the cosine similarity formula:

$$\cos(\theta) = \frac{76}{\sqrt{91} \cdot \sqrt{194}} \approx 0.57$$

The intuition is to take the Frobenius inner product, which measures the alignment of the matrices, and normalize it by the product of their magnitudes (Frobenius norms). This gives us a measure of how similar the two matrices are in terms of their direction in the vector space, regardless of their magnitude. A cosine similarity of 1 means the matrices are perfectly aligned, 0 means they are orthogonal, and –1 means they are opposite.

Summary

Geometrically, all $\mathbb{R}^{m \times n}$ matrices form vector space $\mathbb{R}^{m \cdot n}$

The Frobenius norm is the length of a matrix in this vector space. The Frobenius inner product is the dot product in this vector space. The cosine similarity is the cosine of the angle between two matrices in this vector space.